Phạm Thanh Lâm

Network Engineer · Computer Networks & Data Communications

Network Engineer focused on designing and validating complete solutions across network, security, and cloud environments. Experienced in building real-world systems end-to-end. Seeking Network/Security Engineering roles to contribute to practical system design and operations.

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EXPERIENCE

FSI Technology Joint Stock Company — Mar 2026 - Jun 2026

Network & Security Engineer Intern

26/22 Street No. 03, Do Thanh Residential Area, Ban Co Ward, Ho Chi Minh City

- Validated and deployed **F-DDH BOX** security appliance in lab and production-like environments, performing log analysis, alert verification, and network traffic testing.
- Supported enterprise network infrastructure operations for internal users, including LAN connectivity troubleshooting, endpoint network configuration, shared resource access, and device onboarding.
- Assisted with Cisco network initiatives including **Catalyst 9200L** upgrade research, StackWise study, and SD-WAN solution evaluation; developed technical and presales documentation to support customer demos and enterprise advisory.
- Conducted security research on VPN evasion techniques (**Tailscale, Cloudflare WARP, WireGuard on AWS EC2**) to assess NGFW visibility limitations and security controls.

PROJECTS

HA Firewall & AI-driven Security Monitoring — Graduation Thesis — Jan - May 2026

- Deployed **Active-Passive OPNsense cluster** (CARP/pfsync) with HAProxy SSL offloading and **ModSecurity WAF** (OWASP CRS); **2.8s failover (<5s target)**, zero-session-drop with pfsync, and only **0.71%** sync overhead.
- Executed WAF validation with Nikto scan: **12,796 alerts**, DR 100% / FPR 0% on the baseline payload set; conducted payload-level analysis to quantify security vs. false-positive trade-offs.
- Built a **Wazuh SIEM** pipeline and trained Random Forest / Logistic Regression on a custom attack dataset (DVWA + Kali Linux): average detection time **1.80s**, RF AUC **1.000** (lab), LR AUC **0.981**.

`OPNsense` `CARP/pfsync` `HAProxy` `ModSecurity` `OWASP CRS` `Wazuh SIEM` `Random Forest` `Python` `VMware`

Cisco SD-WAN Hospital Network — Cisco dCloud Lab — Apr 2026

- Architected a 3-site **Hub-and-Spoke SD-WAN overlay** on Cisco dCloud CML (2× C8000v); enforced **IPsec AES-256-GCM across both transports (MPLS + biz-internet)** — guaranteeing patient data is never transmitted in plaintext, even during transport failover.
- Implemented **least-privilege traffic segmentation** via AAR policy: pinned critical medical traffic (Oracle, MS-SQL) exclusively to MPLS, fully isolated from the internet-bound path; verified failover exposes no sensitive traffic through vManage Events Log — **~3 min convergence, zero manual intervention**.

`Cisco CML` `Catalyst SD-WAN 20.18` `vManage` `C8000v` `AAR Policy` `IPsec` `BFD` `Hub-and-Spoke`

AI Shield Firewall — LLM Traffic Inspection Proxy — Personal Project — 2025 - 2026

- Built a **5-layer bilingual (EN+VI) LLM security proxy** on AWS (FastAPI, Docker) enforcing DLP at enterprise AI API perimeters: Regex → Presidio NLP + **Vietnamese PII recognizers (CCCD, tax code, license plate)** → Prompt Injection detection → per-org Policy Engine → Output Inspection; **25/25 unit tests passed**, <15ms proxy overhead, IAM Role credentials + DynamoDB audit logging.

`Python` `FastAPI` `AWS EC2` `Docker` `Presidio NLP` `Prompt Injection` `IAM Role` `DynamoDB`

AutoCTI — Autonomous Cyber Threat Intelligence Agent — Personal Project — 2025 - 2026

- Built a **24/7 automated SOC threat-intelligence pipeline** on AWS Free Tier (~$0/month): **LangGraph ReAct Agent** + Gemini 2.5 Flash + Tavily real-time search for autonomous threat triage and Telegram alerting; IAM Role credentials, DynamoDB conversation memory, and production-grade reliability achieving **~80% RAM reduction** over prior Playwright-based architecture.

`AWS` `SQS` `LangGraph` `Python` `Docker` `Gemini AI` `Telegram Bot API`

CERTIFICATIONS & LEARNING

Networking Basics & Networking Devices — In progress
Cisco Networking Academy

Endpoint Security & Network Defense — In progress
Cisco Networking Academy

EDUCATION

University of Information Technology (UIT – VNU-HCM)

B.Sc. in Computer Networks and Data Communications

GPA 3.09 / 4.0 · 2022 - 2026

SKILLS

NETWORKING

TCP/IP, Routing/Switching, SD-WAN, High Availability, Cisco IOS, OSPF, BGP, VLAN/STP, ACL/NAT, BFD, CARP/pfsync, IPv6

SECURITY

OPNsense, Wazuh SIEM, AI Security, ModSecurity WAF, Suricata IDS/IPS, HAProxy, Nmap, Zabbix

CLOUD & INFRA

AWS, Docker, Linux, VMware, Cisco CML

DEV & TOOLS

Python, C++, FastAPI, Git, REST API, Shell Script, Wireshark

LANGUAGES

Vietnamese — Native
English — Technical reading